Joel James

es24btech11017@iith.ac.in | jamesjoel9705@gmail.com | +91 8089227146 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.23
ASSISI VIDYANIKETAN PUBLIC SCHOOL	Class XII (CBSE)	2023	95.40%
SACRED HEART SCHOOL	Class X (IGCSE)	2021	8.80%

SKILLS

Languages : C, C++, Python

Tools: Git, GNU Make, Linux(WSL)

PROJECTS

Chess Engine (C++)

- Built a high-performance chess engine from scratch featuring optimized move generation, efficient make/undo move execution, and complete support for castling, en passant, promotion, threefold repetition, and the fifty-move rule.
- Implemented alpha-beta pruning, quiescence search, transposition tables, Zobrist hashing, and move-ordering heuristics (MVV-LVA, killer moves, and history heuristic) to significantly improve search efficiency.
- Designed a handcrafted evaluation function incorporating material balance, piece-square tables, mobility, and positional heuristics.
- Achieves approximately **1 million nodes/second** in single-threaded search and recorded a **101-107** match record against **Stockfish Skill Level 3** in automated CuteChess testing (estimated playing strength of **~1500 Elo**).

Jalloc (C)

- Built a custom memory allocator implementing malloc() and free() using the Windows VirtualAlloc API for low-level virtual memory management.
- Implemented free-list allocation with block splitting, constant-time coalescing, multi-segment heap growth, packed metadata headers, and 16-byte aligned allocations.
- Designed a compact chunk layout to minimize metadata overhead while improving memory reuse and reducing external fragmentation.
- Benchmarked allocator throughput and fragmentation against the Windows system allocator across diverse allocation workloads.

C-like Compiler (C)

- Developing a compiler for a subset of C, implementing lexical analysis, recursive parsing, abstract syntax tree (AST) construction, and x86-64 assembly code generation.
- Built a lexer supporting integer literals, identifiers, keywords, and arithmetic operators with lexical error reporting.
- Implemented a Pratt parser to construct ASTs while correctly handling operator precedence and expression parsing.
- Developed a backend that traverses the AST to generate x86-64 assembly with register allocation for arithmetic expressions and print statements.

Custom C++ Container Library

- Developed an STL-inspired container library featuring generic implementations of Vector, Deque, Heap, and Binary Search Tree using C++ templates.
- Implemented custom memory management, move semantics, copy semantics, iterators, and amortized allocation strategies for efficient container operations.
- Designed reusable, type-safe container interfaces emphasizing performance, modularity, and low-overhead abstractions.
- Benchmarked custom containers against the C++ Standard Library to analyze throughput, scalability, and memory efficiency.

RELEVANT COURSEWORK

Computer Science : Intro to Programming, Intro to AI, Operating Systems, Data Structures and Applications, Algorithms, Theory of Computation

Mathematics : Calculus, Elementary Linear Algebra, Numerical Methods, Discrete Maths for CS, Intro to Statistics, Probability and Random Variables

POSITIONS OF RESPONSIBILITY

Class Representative | Engineering Science , IIT Hyderabad | 2024 - 2025

- Acted as the liaison between students and faculty, facilitating communication and resolving academic concerns.

Volunteer | Office of Career Services, IIT Hyderabad | 2024 - 2025

- Supported placement and internship activities.